



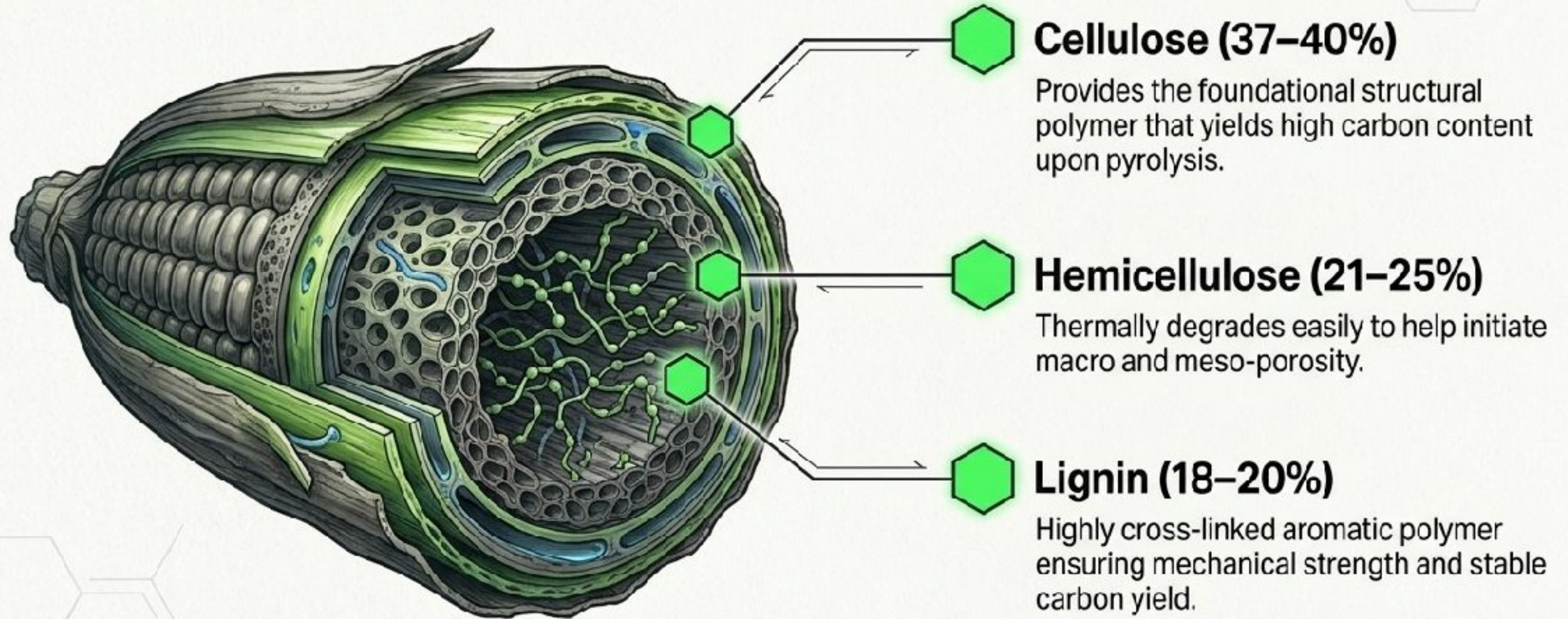
Engineering Clean Water from Agricultural Waste

Synthesizing highly efficient,
economically superior activated
carbon from lignocellulosic
biomass

The Intersection of Two Global Crises



The Lignocellulosic Advantage of Corn Residue

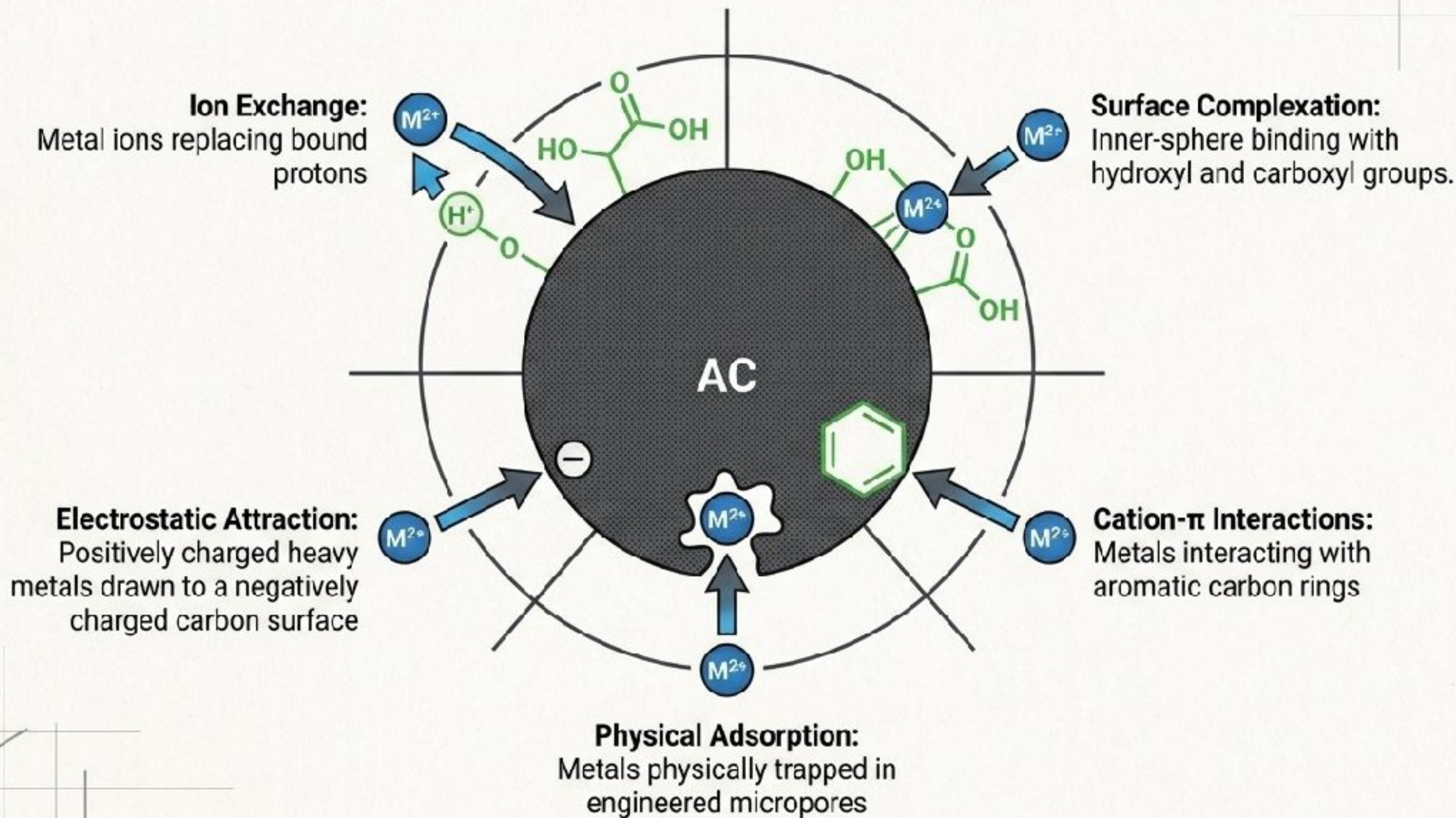


This specific natural makeup produces highly porous carbon materials with immense specific surface areas when proper

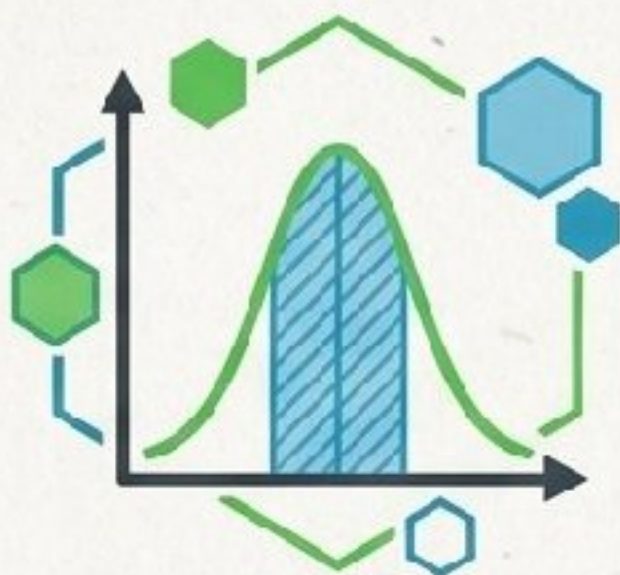
The Transformation Pipeline: Biomass to Active Sorbent



The Anatomy of Heavy Metal Adsorption

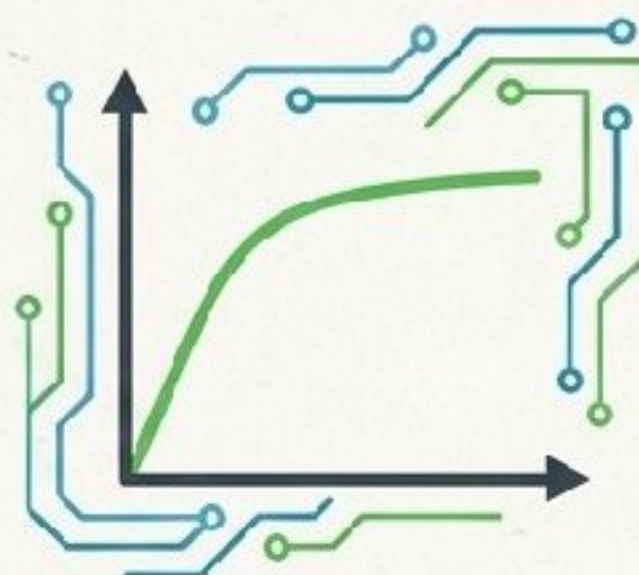


Variables Dictating Adsorption Efficiency



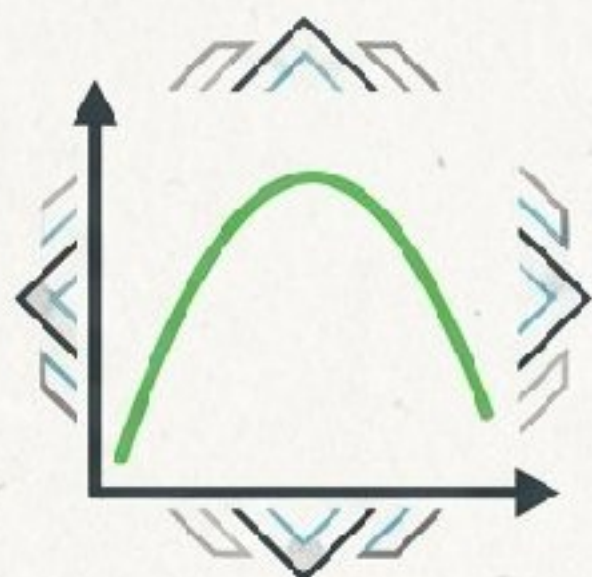
Solution pH [The Master Variable]

Low pH causes H^+ competition and repulsion. Optimal pH yields negative surface charge for cation attraction. Excessive pH causes metal precipitation.



Contact Time

Rapid initial uptake due to abundant surface sites, followed by a slower plateau as intraparticle diffusion limits the rate toward equilibrium.



Adsorbent Dosage

Efficiency scales with mass until a critical threshold where particle agglomeration causes active sites to overlap, reducing effective surface area.



Thermodynamics

Endothermic and spontaneous (Negative ΔG^0 , Positive ΔH^0). Higher temperatures increase randomness at the solid-liquid interface (Positive ΔS^0) improving uptake.

Proof of Concept: Targeting Lead (Pb^{2+})

Experiment Profile

The Threat: Lead Nitrate ($\text{Pb}(\text{NO}_3)_2$)

A highly toxic, non-biodegradable contaminant causing severe nervous system and renal damage.

The Sorbent: Corncob Activated Carbon

Activated via HNO_3 (Nitric Acid) to massively enhance porosity and introduce active binding sites, then neutralized to a stable pH 7.

The Parameters: Controlled Batch Mode

Fixed temperature of 30°C , constant shaking speed of 150 rpm.

The Objective: Performance Optimization

Find the exact mathematical sweet spot for initial concentration, contact time, and dosage to maximize removal efficiency.

Optimization Dashboard: >97% Removal Efficiency

>97.5% Lead Removal achieved under optimal conditions.



Contact Time

Optimal at 90 mins
(Equilibrium reached before desorption begins)

Initial Concentration

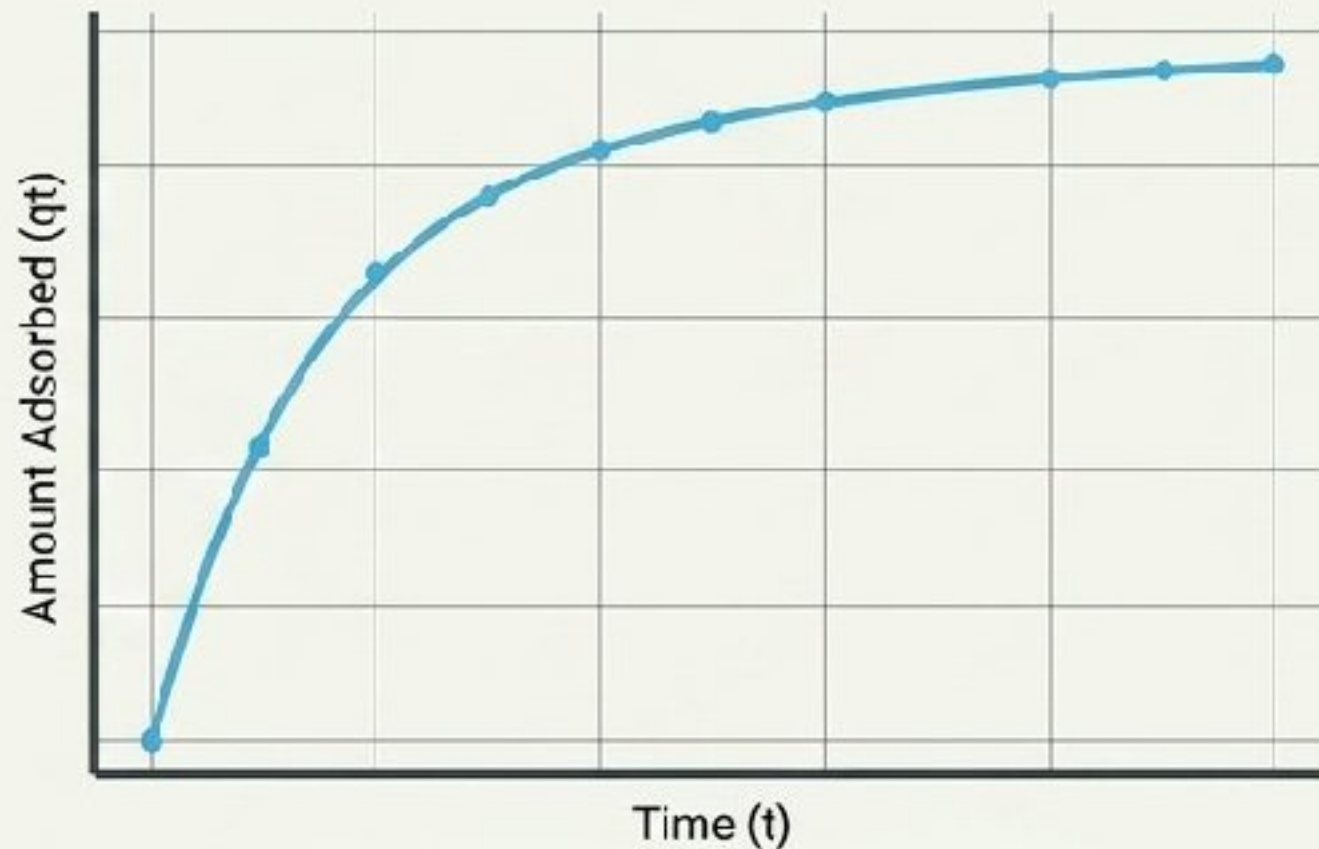
Optimal at 200 mg/L
(Provides ideal concentration gradient driving force)

Adsorbent Dose

Optimal at 0.35 g/L
(Maximizes surface area before particle agglomeration occurs)

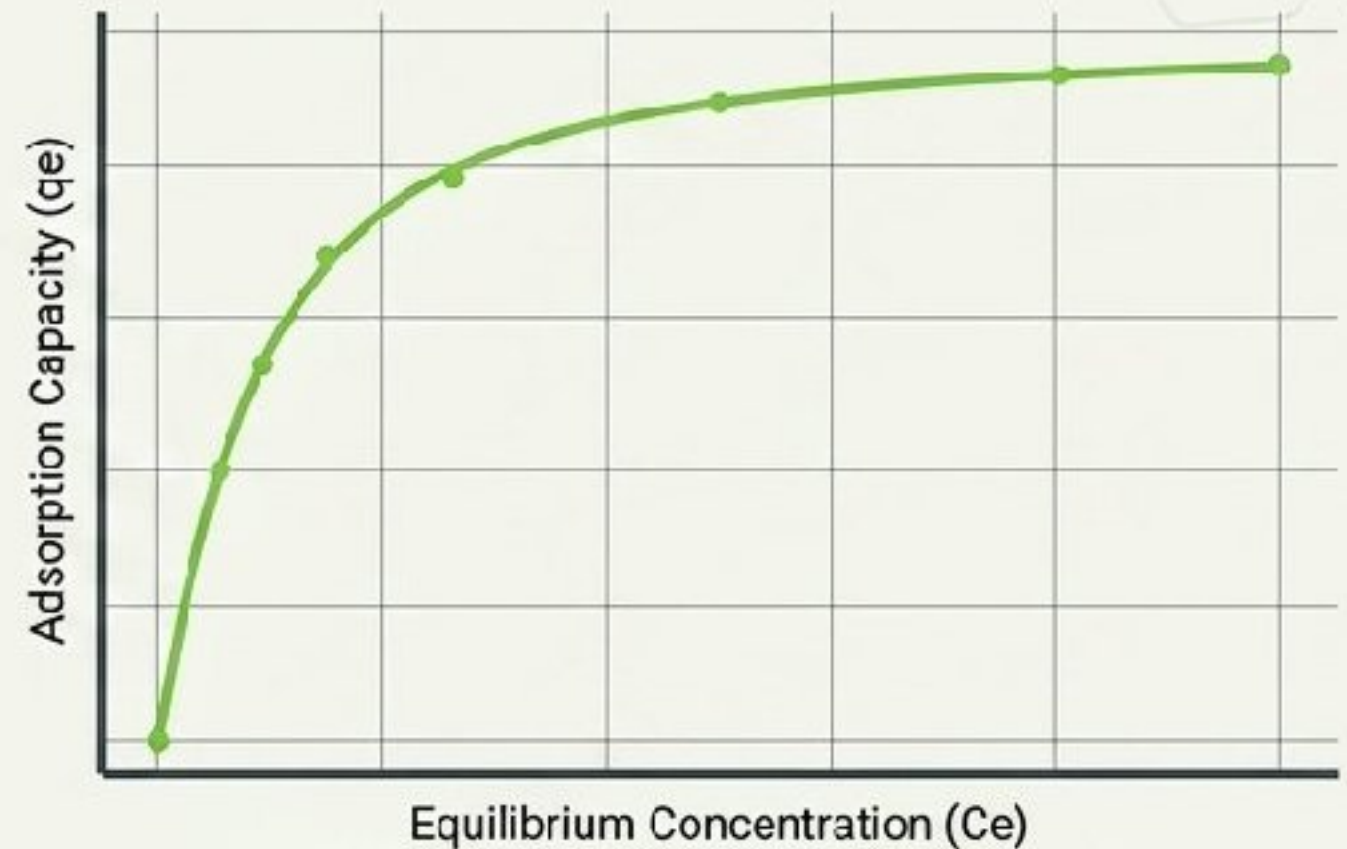
The Mathematical Reality: Proving Chemisorption

Kinetics



The data perfectly fits the pseudo-second-order model ($R^2 \approx 0.99$). This proves the rate-limiting step is chemisorption—the lead is chemically bonding to the surface functional groups, not just getting physically stuck.

Isotherms



Monolayer adsorption capacity models demonstrate that once the surface active sites are filled, equilibrium is maintained, allowing for highly predictable industrial scaling.

Multi-Metal Versatility

Cu

Copper

Up to 643.92 mg/g

Cr

Chromium

Up to 175.44 mg/g

Cd

Cadmium

Up to 87.72 mg/g

Pb

Lead

Up to 206.01 mg/g

Hg

Mercury

Up to 227.32 mg/g

Note: Exact maximum adsorption capacities depend heavily on the specific chemical activation agent (e.g., KOH, Citric Acid, KMnO_4) and the target wastewater matrix.

The Sorbent Showdown: Corncob vs. Convention

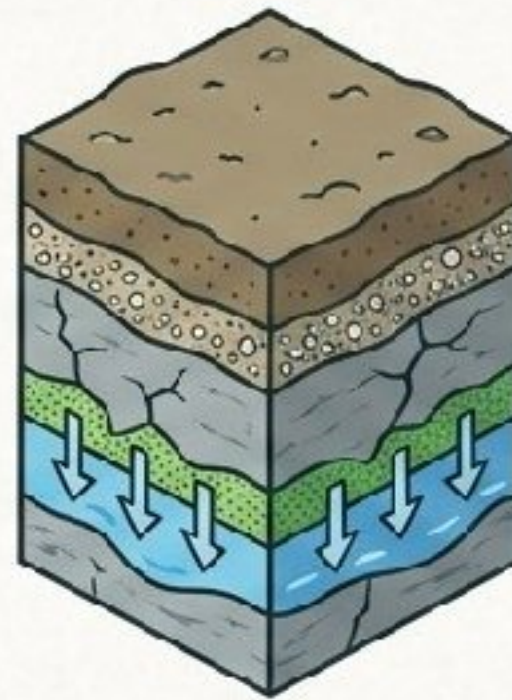
	Material Cost	Adsorption Efficiency	Regeneration Potential	Environmental Footprint
Corncob Activated Carbon	 Agricultural waste	 Highly competitive	 Viable with dilute acids	 Carbon negative
Commercial Activated Carbon	 Highly expensive	 Excellent	 Energy intensive	 High energy production
Ion Exchange Resins	 Very High	 High selectivity	 Highly viable	 Chemical fouling risks
Zeolites	 Moderate	 Lower for some metals	 Moderate	 Mined material

Deploying the Technology in the Real World



Industrial Wastewater

- Treating effluent from battery manufacturing and electroplating.
- Pre-discharge filtration for metallurgy pipelines.



Groundwater Remediation

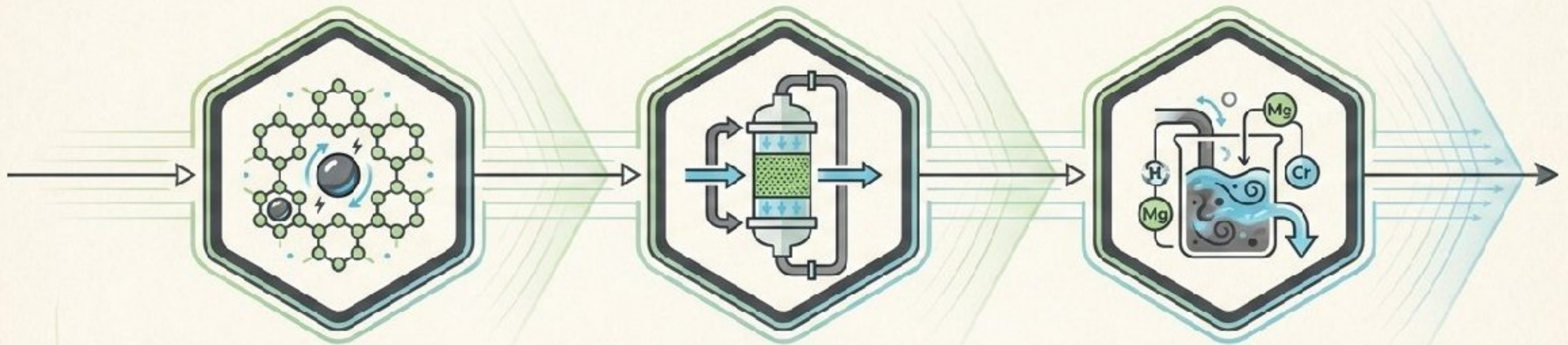
- Filtering aquifers contaminated by landfill leachate.
- Reversing damage from historical mining runoff.



Drinking Water Purification

- Incorporation into low-cost filtration cartridges.
- Specifically engineered for rural and economically vulnerable communities.

Future Frontiers in Bio-Sorbent Engineering



Nanocomposite Integration

Embedding magnetic nanoparticles (e.g., Fe_3O_4 or Ag-SiO_2) directly into the carbon matrix to allow incredibly easy magnetic separation and recovery post-treatment.

Continuous-Flow Systems

Scaling up from laboratory batch experiments to industrial-grade continuous-flow fixed-bed columns for mass water processing.

Multi-Metal Real-World Testing

Moving beyond synthetic laboratory water to evaluate performance against the complex, multi-contaminant realities of actual industrial effluents.

The Circular Economy Realized



Executive Summary: The Corncob Advantage



Highly Effective

Demonstrates massive adsorption capacities across a spectrum of deadly heavy metals, driven by engineered microporosity and rich surface chemistry.



Economically Superior

Transforms a globally abundant, near-zero-cost agricultural nuisance into a premium industrial material, outperforming expensive commercial alternatives.



Environmentally Necessary

Simultaneously mitigates air pollution from biomass burning while securing the future of clean water through sustainable green chemistry.